Samuel E. O'Donnell, PhD

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Research Interests

New particle formation; secondary organic aerosol formation; air quality; climate change; atmospheric modeling; aerosol-cloud interactions; methane emissions; greenhouse gas emissions.

Education

Colorado State University	PhD. Atmospheric Science Advisor: Dr. Jeffrey R. Pierce	2025
Colorado State University	M.S. Atmospheric Science Advisor: Dr. Jeffrey R. Pierce	2022
UNC Asheville	B.S. Atmospheric Science Minor: Mathematics	2018

Research Experience

2025 - current — Postdoctoral Fellow
Topic: Methane emissions modeling from oil and gas facilities
Colorado State University, Department of Systems Engineering
Advisor: Dr. Anna Hodshire

2019 - 2025 — Graduate Research Assistant
Topic: New particle formation and growth in the atmosphere
Colorado State University, Department of Atmospheric Science
Advisor: Dr. Jeffrey R. Pierce

2017 - 2018 — Independent Undergraduate Research
Analysis of ozone concentrations in Houston TX during hurricane Harvey
UNC Asheville, Advisor: Dr. Christopher Hennon

2016 - 2018 — Undergraduate Research Assistant
Vortex-Southeast – *post-tornado damage surveys and GIS analysis*
UNC Asheville, Dr. Christopher Godfrey

2015 - 2018 — Undergraduate Research Assistant
Duke GSMRGN – *rain gauge calibration and data collection*
UNC Asheville, Dr. Douglas Miller

Teaching Experience

Teaching Assistant: **ATS772**, Aerosol Physics, Clouds, and Climate; Jeffrey R. Pierce
ATS555, Air Pollution; Jeffrey R. Pierce

Lecturer: **ATS772**, Aerosol Physics, Clouds, and Climate.

Professional Service

American Association for Aerosol Research CSU chapter – president

Served as the president of the CSU AAAR chapter from 2022-2025.

Micro2Macro Breakout Session Lead

Served as the lead for the Aerosol Chemical and Physical Properties breakout session during the 2024 Micro2Macro workshop.

SOARS protégé – panelist

Served as a panelist for a discussion with SOARS protégé program.

CIRA/ATS Mentoring Program (CAMP) – mentor

Served as a graduate mentor for first year graduate students in the CSU ATS department.

Promoting Geoscience Research, Education & Success (PROGRESS) mentoring program – mentor

Served as a mentor for program supporting undergraduate women in STEM

CSU Department of Atmospheric Science Code of Conduct – author.

Wrote the initial draft and helped propose a Code of Conduct for the CSU atmospheric science department.

Little Shop of Physics

Volunteer for the Little Shop of Physics at CSU.

Reviewer – Environmental Science and Technology, Journal of Geophysical Research

Publications

Submitted:

1. Regina Luu, Meredith Schervish, **Samuel E. O'Donnell**, Jeffrey R. Pierce, and Manabu Shiraiwa: Kinetic Partitioning with Effective Mass Accommodation for Secondary Organic Aerosol Simulations in GEOS-Chem, *submitted to Geophys. Mod. Dev.*, 2026.

Published:

1. Nicole A. June, Cort L. Zang, **Samuel E. O'Donnell**, Ben G. Kopec, Anderson Da Silva, Betty Croft, Rachel Y.-W. Chang, Linia Tashmim, Antoine Haddon, Shengqian Zhou, William C. Porter, Jeffrey M. Welker, Shantanu H. Jathar, Nadja Steiner, Michael Tjernström, Paul Zieger, Megan D. Willis, Jeffrey R. Pierce: Through the Cracks: Sea Ice Regions as a Source of Volatile Marine Sulfur in the Springtime Arctic, in press at Journal of Geophysical Research, 2026.

2. En Li, **Samuel E. O'Donnell**, Adam T. Ahern, Kelvin H. Bates, Charles A. Brock, Wyndom S. Chace, Matthew M. Coggon, Emily V. Fischer, Jessica B. Gilman, Georgios I. Gkatzelis, Han N. Huynh, Emily R. Lill, Ming Lyu, Ann M. Middlebrook, Alison M. Piasecki, Jeff Peischl, Ilana B. Pollack, Andrew W. Rollins, Nell B. Schafer, Chelsea E. Stockwell, Victoria Treadaway, Carsten Warneke, Eleanor M. Waxman, Lu Xu, Kristen Zuraski, Robin G. Stevens, Shantanu H. Jathar, Jeffrey R. Pierce: Sulfur Dioxide Point Sources Remain Important for Daytime Plume-Scale New Particle Formation in the United States: Case Study for Gary, Indiana during the Summer 2023 AEROMMA Aircraft Campaign, *Journal of Geophysical Research*, in press, 2026.

3. **Samuel E. O'Donnell**, Xuanlin Du, Jeremy Wakeen, Abraham Dearden, Yicong He, Rebecca J. Sheesley, James N. Smith, Don R. Collins, Shantanu H. Jathar, Jeffrey R. Pierce: Case study probing the potential drivers of Houston new particle formation and growth in a unique outdoor chamber, *Aerosol Science & Technology*, https://doi.org/10.1080/02786826.2026.2669561, 2026.

4. Jeremy Wakeen, **Samuel E. O'Donnell**, Don R. Collins, Jeffrey R. Pierce, James N. Smith: Positive matrix factorization reveals volatility-resolved composition from new particle formation

during α-pinene ozonolysis, *Aerosol Science & Technology*, https://doi.org/10.1080/02786826.2026.2649824. 2026.

5. **Samuel E. O'Donnell**, Betty Croft, Bonne Ford, Nicole A. June, Chongai Kuang, Ashish Singh, Rachel Y.-W. Chang, Don R. Collins, Simo Hakala, Shantanu H. Jathar, Pauli Paasonen, Manish Shrivastava, James N. Smith, Jeffrey R. Pierce; Going Off Grid: A Comparative Study of the Lagrangian and Eulerian Perspectives of New Particle Formation Events, *J. Geophys. Res.,* https://doi.org/10.1029/2025JD043713, 2025.

2. Regina Luu, Meredith Schervish, Nicole June, **Samuel E. O'Donnell**, Shantanu Jathar, Jeffrey Pierce, Manabu Shiraiwa; Global Simulations of Phase State and Equilibration Timescales of Secondary Organic Aerosols with GEOS-Chem, *ACS Earth and Space Chemistry*, https://doi.org/10.1021/acsearthspacechem.4c00281, 2025.

3. Manish Shrivastava, Jie Zhang, Rahul A. Zaveri, Bin Zhao, Jeffrey R. Pierce, **Samuel E. O'Donnell**, Jerome Fast, Brian Gaudet, John Shilling, Alla Zelenyuk, Ben Murphy, Havala Pye, Qi Zhang, Justin Trousdell, Renyi Zhang, Yixin Li, Qi Chen; Anthropogenic extremely low volatility organics (ELVOCs) Govern the Growth of Molecular Clusters over the Southern Great Plains during the Springtime, *J. Geophys. Res.*, http://dx.doi.org/10.1029/2024JD041212, 2024.

4. **Samuel E. O'Donnell**, Ali Akherati, Yicong He, Anna L. Hodshire, John E. Shilling, Chongai Kuang, Jerome D. Fast, Fan Mei, Siegfried Schobesberger, Joel A. Thornton, James N. Smith, Shantanu H. Jathar, Jeffrey R. Pierce; Look up: Probing the vertical profile of new particle formation and growth in the planetary boundary layer with models and observations*, J. Geophys. Res.*, http://doi.org/10.1029/2022JD037525, 2023.

5. **Samuel E. O'Donnell**; Case Study Analysis of Ozone Concentrations in Houston TX During Hurricane Harvey, *UNC Asheville J. Undergrad. Res.*, 2018, available at: [https://sites.google.com/a/unca.edu/unc-asheville-journal/2018/search-by-author-2018?authuser=0]

Presentations

O'Donnell, S. E., Zimmerle, D., Hodshire, A. L; *Informed decisions; a review of how inventories are informed by measurements*. EEMDL, **2026**.

O'Donnell, S. E., Du, X., Wakeen, J., Smith, J. N., Collins, D. R., He, Y., Dearden, A., Jathar, S. H., Pierce, J. R; *Probing the Potential Drivers of New Particle Formation and Growth During TRACER in Houston, TX.*, 105th AMS Annual Meeting, **2025**.

O'Donnell, S. E., Ford, B., Croft, B., June, N. A., Smith, J. N., Collins, D. R., Shrivastava, M., Paasonen, P., Jathar, S. H., Pierce, J. R; *Going Off Grid: A Comparative Study of the Lagrangian and Eulerian Perspectives of New Particle Formation Events*, Micro2Macro: Origins of Climate Uncertainty, **2024**.

O'Donnell, S. E., Ford, B., Croft, B., June, N. A., Smith, J. N., Collins, D. R., Shrivastava, M., Paasonen, P., Jathar, S. H., Pierce, J. R; *Understanding New Particle Formation Events in GEOS-Chem by Comparing the Lagrangian and Eulerian Perspectives*, 11th International GEOS-Chem Meeting, **2024**.

O'Donnell, S. E., Ford, B., Croft, B., June, N. A., Jathar, S. H., Pierce, J. R; *Explaining Stationary Observations of New-Particle Formation and Growth Using a Lagrangian Perspective*, American Geophysical Union Annual Conference, **2023**.

O'Donnell, S. E., Croft, B., Jathar, S. H., Pierce, J. R.; *Lagrangian Versus Eulerian Perspectives of New Particle Formation Events*, The 21st International Conference on Nucleation & Atmospheric Aerosols, **2023**.

O'Donnell, S. E., Akherati, A., He, Y., Hodshire A. L., Shilling J. E., Kuang C., Fast J. D., Mei F., Schobesberger S., Thornton J. A., Smith J. N., Jathar S. H., Pierce J. R.; *Look Up: Probing the Vertical Profile of New Particle Formation and Growth in the Planetary Boundary Layer Using a Column Model with Surface- and Aircraft-Based Observations*, American Association for Aerosol Research 40th annual conference, **2022**.

O'Donnell, S. E., Akherati, A., He, Y., Hodshire A. L., Shilling J. E., Kuang C., Fast J. D., Mei F., Schobesberger S., Thornton J. A., Smith J. N., Jathar S. H., Pierce J. R.; *Process Level Modeling of Vertically Resolved New-particle Formation at the Southern Great Plains Observatory*, International Aerosol Modeling Algorithms Conference, **2021**.

O'Donnell, S. E., Akherati, A., He, Y., Hodshire A. L., Shilling J. E., Kuang C., Fast J. D., Mei F., Schobesberger S., Thornton J. A., Smith J. N., Jathar S. H., Pierce J. R.; *Process Level Modeling of Vertically Resolved New-particle Formation at the Southern Great Plains Observatory*, American Association for Aerosol Research 39th Annual Conference, **2021**.

O'Donnell, S. E., Akherati, A., He, Y., Hodshire A. L., Shilling J. E., Kuang C., Fast J. D., Mei F., Schobesberger S., Thornton J. A., Smith J. N., Jathar S. H., Pierce J. R.; *Towards understanding the vertical profile and drivers of new-particle formation at the DoE Southern Great Plains field site*, AMS 101st Annual Meeting: 23rd Conference on Atmospheric Chemistry, **2021**.

O'Donnell, S. E.; *Case Study Analysis of Ozone Concentrations in Houston TX During Hurricane Harvey*, UNC Asheville Undergraduate Research Symposium, **2018**.

Technical Skills

- Python
  - Scientific data analysis using `numpy, pandas, scipy, xarray`.
  - Data visualization and plotting using `matplotlib` and `seaborn`.
  - Basic machine learning with `Tensorflow` and `scikit-learn` from machine learning courses.
- FORTRAN
  - Worked with and heavily modified several `FORTRAN` based code bases.
  - Developed an Introduction to `FORTRAN` document for the research group.
- MATLAB
  - Data analysis of new particle formation events and simple aerosol models.
- Shell command line – `bash, cmake, slurm`.
- High performance computing using CSU's Asha cluster.
- Atmospheric Modeling:
  - GEOS-Chem with TwO-Moment Aerosol Sectional (TOMAS) microphysics package;
  - Statistical Oxidation Model (SOM)-TOMAS;
  - HYSPLIT.
- Version control using Git and GitHub.

Awards and Honors

2025	2025 **Outstanding Reviewer**, Geophysical Research Letters.
2025	Wiley, Journal of Geophysical Research: Atmospheres, **Top Viewed Article**
2023	David L. Dietrich Honorary Scholarship – CSU. Awarded to a graduate student who has excelled in the study and research of air quality.
2022	**Student poster award** – American Association for Aerosol Research
2021	**First place** – American Meteorology Society student poster presentation.
2018	**Excellence in Research** – Department of Atmospheric Sciences, UNC Asheville
2018	Undergraduate Research Scholar – UNC Asheville